ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 2

MECHATRONICS

PAPER 1 · PRACTICE G

MARKS: 100 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 2. Laid out like the DHET Mechatronic Systems paper: FIVE questions of 20 marks. Levers, pulleys, mechanical advantage, drives and bearings, relay logic and truth tables, compressors and cylinder speed control, relays, contactors and multimeters, and lubrication, condition checks and fault-finding methods.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 2. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of FIVE questions. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Show ALL calculations. Draw all symbols and circuits neatly, using a ruler.
5. Use only a black or blue pen. Write neatly and legibly.

QUESTION 1: MECHANICAL SYSTEMS

1.1 Name the THREE classes of levers and give ONE example of each. (6)

1.2 A lever lifts a load of 600 N with an effort of 150 N. Calculate the mechanical advantage. (2)

1.3 Give TWO advantages of a chain drive over a belt drive. (4)

1.4 Explain the function of a bearing, and name TWO types of bearings. (4)

1.5 A pulley system has a velocity ratio of 4. Explain what this means. (4)

[20]

QUESTION 2: LOGIC AND RELAYS

2.1 Draw a circuit in which a normally closed contact of relay K1 switches lamp E1, so that the lamp is ON when K1 is not energised (NOT function). (4)

2.2 Draw the truth table of a three-input AND gate with inputs A, B and C and output Q. (5)

2.3 Explain how a relay can latch (hold itself on) after a START button is released. (4)

2.4 Explain the difference between a relay and a contactor. (4)

2.5 Write the following in full: AC; NO; NC. (3)

[20]

QUESTION 3: PNEUMATICS

3.1 Name TWO types of air compressors. (2)

3.2 Explain why moisture must be removed from compressed air. (3)

3.3 Explain how the speed of a double-acting cylinder can be controlled, and draw the symbol of the valve used. (6)

3.4 Draw the symbol of EACH of the following: (6)

3.4.1 3/2-way valve, roller operated, spring return (4)

3.4.2 Pressure gauge (2)

3.5 What is a quick-exhaust valve used for? (3)

[20]

QUESTION 4: ELECTRICAL COMPONENTS AND MEASUREMENT

4.1 Draw the symbol of a relay coil and of its normally open and normally closed contacts. (4)

4.2 Explain how a relay works. (4)

4.3 Explain how a multimeter is connected to measure (a) the voltage across a lamp and (b) the current through the lamp. (4)

4.4 Explain why the resistance of a component must never be measured while the circuit is switched on. (2)

4.5 Give THREE reasons why electric motors are widely used in mechatronic systems. (3)

4.6 State the colours used for the live, neutral and earth conductors in a South African three-core cable. (3)

[20]

QUESTION 5: MAINTENANCE AND FAULT-FINDING

5.1 Give FOUR reasons why moving parts must be lubricated. (4)

5.2 Name FOUR signs that a motor or bearing may be about to fail. (4)

5.3 Explain the input-to-output method of fault-finding. (4)

5.4 Give FOUR safety rules for working with compressed air. (4)

5.5 Explain why a maintenance history (logbook) is kept for every machine. (4)

[20]

TOTAL: 100

END OF QUESTION PAPER